IX. Multiple Random Variables (GSE 518)

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Idea: In practice, we usually work with more than one random variable. Moreover, we are usually interested in modeling or learning about one random variable *conditional* (or as a function) of others. Hence, we need concepts about *multiple* random variables.

1. Multiple Random Variables: Introduction

Definition: An *n-dimensional random vector* $X = (X_1, X_2, \ldots, X_n)$ is a function from S to $\mathbb{R}^n$.

Notes

- For simplicity, we focus on a bivariate vector $\{X, Y\}$.
- Example: Let $S = \{\text{Cal Poly students}\}$. define the random vector:

$$\{X(s), Y(s)\} = \{\text{weight of student } s,\ \text{height of student } s\}$$

- Initial concepts to discuss:
  - Joint cdf
  - Joint and marginal pmf/pdf
  - Expectation of $g(X, Y)$

2. Distribution Functions and Expectations

2.1. Joint Cumulative Distribution Function (cdf)

Definition: The *joint cdf* of (X, Y) is

$$F_{XY}(x, y) = P(X \leq x \text{ and } Y \leq y), \quad \text{for all values } (x, y) \in \mathbb{R}^2.$$

Notes on Joint cdf:

- It's defined $\forall (x, y) \in \mathbb{R}^2$
- Same properties as before for cdf's. For example:

$$\lim_{x \to \infty, y \to \infty} F_{XY}(x, y) = 1 \quad \text{and} \quad \lim_{x \to -\infty, y \to -\infty} F_{XY}(x, y) = 0$$

2.2. Joint and Marginal Probability Mass Functions (pmf) and Probability Density Functions (pdf)

a) Case 1: Discrete Random Variables

Definition: The *joint pmf* of (X, Y) is the function $f_{XY}(x, y) : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f_{XY}(x, y) = P(X = x, Y = y), \quad \text{for all } (x, y) \in \mathbb{R}^2$$

. Notes on Joint pmf:

- Same pmf properties as before: $0 \leq f_{XY}(x, y) \leq 1$.
- $\sum_{x \in \mathbb{R}} \sum_{y \in \mathbb{R}} f_{XY}(x, y) = 1$.
- For any $A \subseteq \mathbb{R}^2$, we have that $P((X, Y) \in A) = \sum_{(x,y)} \in A f_{XY}(x, y)$.

Example: Toss a fair coin 3 times. Let $X(s) = \#$ of heads in first 2 tosses, and $Y(s) = \#$ of heads in last 2 tosses.

(i) Find pmf $f_{XY}(x, y)$.

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$ (each with probability $\frac{1}{8}$).

Then, $(X, Y) = \{(2,2), (2,1), (1,1), (1,0), (1,2), (1,1), (0,1), (0,0)\}$.

Note that (X, Y) maps S to $\mathbb{R}^2$.

Given this information, the pmf of (X, Y) is given by:

$$f_{X,Y}(x,y) = P(X = x, Y = y) = \begin{cases} \frac{1}{8} & \text{if } (0,0) \\ \frac{1}{8} & \text{if } (0,1) \\ \frac{1}{4} = \frac{2}{8} & \text{if } (1,1) \\ \frac{1}{8} & \text{if } (1,2) \\ \frac{1}{8} & \text{if } (1,0) \\ \frac{1}{8} & \text{if } (2,1) \\ \frac{1}{8} & \text{if } (2,2) \\ 0 & \text{otherwise} \end{cases}$$

- (ii) Use $f_{XY}(X, Y)$ to find $Pr(X \geq 1, Y \leq 1)$.

  (Note: See solution from lecture)

Definition: The pmf of X or Y alone are still the same as before, $f_X(x) = P(X = x)$ and $f_Y(y) = P(Y = y)$. In the context of many random variables we call $f_X(x)$ and $f_Y(y)$ the *marginal pmf's* of X and Y, respectively.

-Example (continuation): Find the sample space and pmf of X, $f_X(x)$.

(Note: See solution from lecture)

How to get Marginal pmf's from the Joint pmf

Theorem: *(Thm. 4.1.6 from textbook).* If (X, Y) has joint pmf $f_{XY}(x, y)$, then

$$f_X(x) = \sum_{y \in \mathcal{Y}} f_{XY}(x, y) \quad \text{(to get marginal pmf of } X\text{, sum over all possible values of } Y\text{)}$$

$$f_Y(y) = \sum_{x \in \mathcal{X}} f_{XY}(x, y) \quad \text{(to get marginal pmf of } Y\text{, sum over all possible values of } X\text{)}$$

-Example (continuation): Find $f_X(x)$ using the joint pmf $f_{XY}(x, y)$.

(Note: See solution from lecture)

b) Case 2: Continuous Random Variable

Definition: The function $f_{XY}(x, y) : \mathbb{R}^2 \to \mathbb{R}$ is the *joint pdf* of (X, Y) if it satisfies

$$F_{XY}(x, y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{XY}(x, y)\, dx\, dy$$

Notes on Joint pdf:

- Same properties as before for a pdf. For example, $f_{XY}(x, y) \geq 0$ and

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y)\, dx\, dy = 1$$

- To get joint pdf from cdf

$$f_{XY}(x, y) = \frac{\partial^2 F_{XY}(x, y)}{\partial x \partial y}$$

- If (X, Y) have joint pdf $f_{XY}(x, y)$, then we can get the *marginal pdf's* of X and Y, respectively $f_X(x)$ and $f_Y(y)$, as:

$$f_X(x) = \int_{y \in \mathcal{Y}} f_{XY}(x, y)\, dy$$

$$f_Y(y) = \int_{x \in \mathcal{X}} f_{XY}(x, y)\, dx$$

- Example: Let $f_{XY}(x, y) = \frac{2}{3}(X + 2Y)$ for $x \in [0, 1]$ and $y \in [0, 1]$ and 0 otherwise.

(i) Check that the pdf $f_{XY}(x, y)$ integrates to 1.

(ii) Find the marginal pdf's of X and Y, respectively $f_X(x)$ and $f_Y(y)$.

(iii) Find $F_{XY}(1/2, 1/4)$

(Note: See solutions from lecture.)

2.3 Expectations

Definition: The *expectation* of a function $g(X, Y)$ is:

$$E[g(X,Y)] = \begin{cases} \int\limits_{y \in \mathcal{Y}} \int\limits_{x \in \mathcal{X}} g(x, y) \cdot f_{XY}(x, y)\, dx\, dy & \text{if } X, Y \text{ are continuous.} \\ \sum\limits_{y \in \mathcal{Y}} \sum\limits_{x \in \mathcal{X}} g(x, y) \cdot f_{XY}(x, y) & \text{if } X, Y \text{ are discrete.} \end{cases}$$

Example 1: Discrete Case (Continuation). For the example in part 2.2.a, find $E(XY)$.

(Note: See lecture for solution.)

Example 2: Continuous Case (Continuation). For the example in part 2.2.b, find $E(XY)$.

$$E(XY) = \int_0^1 \int_0^1 xy \cdot \frac{2}{3}(x + 2y)\, dx\, dy$$

$$= \frac{2}{3} \int_0^1 \int_0^1 \left(x^2 y + 2xy^2\right) dx\, dy$$

$$= \frac{2}{3} \int_0^1 \left(\frac{1}{3}y + y^2\right) dy$$

$$= \frac{2}{3} \left(\frac{1}{6} + \frac{1}{3}\right)$$

$$= \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{3}$$

- Note: We can calculate $E(X)$ as

$$E(X) = \int\limits_{x \in \mathcal{X}} x \cdot f_X(x) dx$$

or

$$E(X) = \int\limits_{y \in \mathcal{Y}} \int\limits_{x \in \mathcal{X}} x \cdot f_{XY}(x, y) dx dy$$

3. Conditional Distributions and Expectations

Idea: Usually we are interested in the value of a random variable given the value of another (e.g., earning given education level).

Recall that if $P(B) > 0$, then $P(A|B) = \frac{P(A \cap B)}{P(B)}$.

3.1 Case 1: Discrete Random Variables

Definition: Given two discrete random variables X and Y with joint pmf $f_{XY}(x, y)$, and $f_X(x)$ the marginal pmf of X, the *conditional pmf* of Y given $X = x$ is:

$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)}$$

Notes on Conditional pmf:

- In $f_{Y|X}(y|x)$, we think of Y as random and X as fixed or given.
- $f_{Y|X}(y|x)$ is a valid pmf $\Rightarrow$ The usual properties of a pmf must hold, such as $0 \leq f_{Y|X}(y|x) \leq 1$ and $\sum_{y \in \mathcal{Y}} f_{Y|X}(y|x) = 1$.

  Careful! Note that we sum over Y because here Y is random and X is given. In general,

  $$\sum_{x \in \mathcal{X}} f_{Y|X}(y|x) \neq 1$$

- Example (Continuation): For the example in part 2.2.a, find the conditional pmf of Y given $X = 2$, or $f_{Y|X}(y|X = 2)$.

  (Note: See lecture for solution.)

Definition: The *conditional expectation* of $g(Y)$ given $X = x$ is:

$$E[g(Y)|X = x] = \sum_{y \in \mathcal{Y}} g(y) f_{Y|X}(y|x)$$

Notes on Conditional Expectation:

- A special case:

  $$E[Y|X = x] = \sum_{y \in \mathcal{Y}} y f_{Y|X}(y|x)$$

- Key: We sum over Y because Y is the random variable and X is given.
- Example (Continuation): For the example in part 2.2.a, find $E(Y|X=2)$.

  (Note: See lecture for solution.)

3.2. Case 2: Continuous Random Variables

Definition: Given two continuous random variables with joint pdf $f_{XY}(x,y)$, and $f_X(x)$ the marginal pdf of X, the *conditional pdf* of Y given $X=x$ is:

$$f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)} \text{ provided } f_X(x) > 0$$

Notes on Conditional pdf:

- $f_{Y|X}(y|x)$ is a valid pdf, so the usual properties apply such as $f_{Y|X}(y|x) \geq 0$ and $\int\limits_{y\in\mathcal{Y}} f_{Y|X}(y|x)\,dy = 1$
- Key: As before, we integrate over Y because Y is the random variable and X is given. If we integrate over X there is no reason for the pdf to integrate to 1.

Definition: The *conditional expectation* of $g(Y)$ given $X=x$ is:

$$E[g(Y)|X=x] = \int\limits_{y\in\mathcal{Y}} g(y) f_{Y|X}(y|x)\,dy$$

Notes on Conditional Expectation (Continuous Case):

- A special case is:
$$E[Y|X=x] = \int\limits_{y\in\mathcal{Y}} y f_{Y|X}(y|x)\,dy$$
- As before, we integrate over Y because Y is the random variable and X is given.

- Example (Continuation): Consider the example from part 2.2.b.

(i) Find the conditional pdf of Y given X, or $f_{Y|X}(y|x)$

(ii) Show that $f_{Y|X}(y|x)$ is a valid pdf (i.e., it is non-negative and it integrates to 1)

(iii) Find $E(Y|X)$

(iv) Find $Pr(Y \leq 1/2|X = 1)$

(Note: See solutions from lecture.)

Key Note: *In all of these problems always keep in mind (i) what pdf do we use (e.g., joint, marginal, conditional)? (ii) Which variable is random and which is thought as fixed? (iii) With respect to which variable do we integrate (or sum) over?*

4. Independence

Recall: If events A and B are independent, then: $P(A \cap B) = P(A) \cdot P(B)$.

Definition: Let the random variables X and Y have joint pmf/pdf $f_{XY}(x, y)$ and marginal pmf's/pdf's $f_X(x)$ and $f_Y(y)$, respectively. Then, X and Y are independent if $\forall (x, y) \in \mathbb{R}^2$

$$f_{XY}(x, y) = f_X(x) \cdot f_Y(y)$$

4.1. Useful Result to Check for Independence

Theorem (Factorization Theorem): *(Lemma 4.2.7. in textbook).* Two random variables X and Y are independent *if and only if* there exists a pair of functions $g(x)$ and $h(y)$ such that the joint pmf/pdf of X and Y can be written as:

$$f_{XY}(x, y) = g(x) \cdot h(y) \quad \forall (x, y) \in \mathbb{R}^2$$

- See examples from lecture.

4.2. Implications of Independence

1. Let X and Y be independent. Then,

$$f_{Y|X}(y|x) = f_Y(y)$$

$$f_{X|Y}(x|y) = f_X(x)$$

2. Let X and Y be independent. Then, for any functions $r(X)$ and $s(Y)$, we have that:

$$E[r(X) \cdot s(Y)] = E[r(X)] \cdot E[s(Y)]$$

   Special case:

$$E[XY] = E[X]E[Y]$$

3. **Theorem:** *(Thm. 4.2.12 from textbook).* If $Z = X+Y$ and X and Y are independent with mgf's $M_X(t)$ and $M_Y(t)$, then the mgf of Z is:

$$M_Z(t) = M_X(t) \cdot M_Y(t)$$

4. **Theorem:** *(Thm. 4.3.5 from textbook).* If X, Y are independent, and we have $U = g(X)$ and $V = h(Y)$, then U and V are independent.

5. Covariance and Correlation

Goal: Define a scalar summary measure of the degree of linear dependence between X and Y.

5.1. Covariance

Definition: The *covariance* of X and Y is:

$$\text{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)], \quad \text{where} \quad \mu_X = E(X) \quad \text{and} \quad \mu_Y = E(Y)$$

Notes on covariance:

- Interpretation:
  - $\text{Cov}(X,Y) > 0$ if on average $Y > \mu_Y$ when $X > \mu_X$ (or vice versa), or $Y < \mu_Y$ when $X < \mu_X$ (or vice versa).
  - $\text{Cov}(X,Y) < 0$ if on average $Y > \mu_Y$ when $X < \mu_X$ (or vice versa), or $Y < \mu_Y$ when $X > \mu_X$ (or vice versa).

    Note: The order of X and Y does *not* matter (thus the "or vice versa" wording).
- Covariance does **not** imply Causation
  - Example: Let $X =$ education $\quad Y =$ earnings.
    - $\text{Cov}(X,Y) > 0$ means on average we observe that education is above its mean when earnings are above its mean, (or $X < \mu_X$ then $Y < \mu_Y$).
    - It does **not** imply "X causes Y" or "Y causes X", or even "X affects Y" or "Y affects X"
    - For $Cov(X,Y)$, the order of X and Y does not matter.
    - It only says, on average, what the behavior of X and Y are relative to their respective means.

Important Covariance Formulas:

a) $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$ *(See "proof" from lecture)*

b) $\mathrm{Cov}(X, Y) = E(XY) - E(X)E(Y) = E(XY) - \mu_X \mu_Y$

c) For a constant a, $\mathrm{Cov}(X, a) = 0$.

d) For constants a, b, c, d and random variables X, Y, U, V:

$$\mathrm{Cov}(aX+bY, cU+dV) = ac\cdot\mathrm{Cov}(X,U)+ad\cdot\mathrm{Cov}(X,V)+bc\cdot\mathrm{Cov}(Y,U)+bd\cdot\mathrm{Cov}(Y,V)$$

- Example: Let $Cov(X, Y) = 4$, $Var(X) = 2$ and $Var(Y) = 3$. Also, let $W = 2X + 4$ and $U = 3X + 2Y$. *Find* $Cov(W, U)$. (See solution from lecture.)

e) Theorem: *(Thm. 4.5.6. in textbook).* For constants a, b we have:

$$\mathrm{Var}(aX + bY) = a^2 \cdot \mathrm{Var}(X) + b^2 \cdot \mathrm{Var}(Y) + 2ab \cdot \mathrm{Cov}(X, Y)$$

- Example (Continuation of previous above): *Find* $Var(U)$.
(See solution from lecture.)

(Note: See the "proof" or "why?" of each formula from lecture.)

5.2. Correlation

Definition: The correlation of X and Y is:

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad \text{where} \quad \sigma_X^2 = \mathrm{Var}(X) \quad \text{and} \quad \sigma_Y^2 = \mathrm{Var}(Y)$$

Notes on Correlation:

a)
- We can think of it as a standardized version of covariance, where we standardize by σ_X and σ_Y.
- The Correlation is easier to interpret than covariance (see below).
- ρ_{XY} does **not** imply causation.

b) $-1 \leq \rho_{XY} \leq 1$

- *Proof:*
  - This proof uses the results previously discussed. To do so, write:

$$0 \leq \text{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right)$$

  - Next, apply part (e) from "Important Covariance Formulas" above to get:

$$0 \leq \frac{Var(X)}{\sigma_X^2} + \frac{Var(Y)}{\sigma_Y^2} + 2\frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

$$0 \leq 1 + 1 + 2\rho_{XY} \implies 0 \leq 2(1 + \rho_{XY}) \implies \boxed{-1 \leq \rho_{XY}}$$

  - To complete the proof, write:

$$0 \leq \text{Var}\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right)$$

  - Then, follow the same steps as before to show $\boxed{\rho_{XY} \leq 1}$

c) $\rho_{XY} = \begin{cases} 1 & \text{if } Y = aX + b \text{ for some } a > 0 \text{ and } b \in \mathbb{R} \\ -1 & \text{if } Y = aX + b \text{ for some } a < 0 \text{ and } b \in \mathbb{R} \end{cases}$

- If $\rho_{XY} = 1$ or -1 then we have *exact* linear dependence between X and Y.
- The closer $|\rho_{XY}|$ is to 1, the stronger the linear dependence between X and Y.

d) ρ_{XY} is unitless while $\text{Cov}(X, Y)$ is unit dependent.

- This is why ρ_{XY} is easier to interpret.
- Example:
  - If we observe that $\text{Cov}(X, Y) = 1000$, it is hard to say whether this implies that the linear relationship between X and Y is strong or weak, since it depends on the units.
  - However, if we observe that $\rho_{XY} = .95$, we can easily say that X and Y have strong linear association.

e) **Theorem:** (*Thm. 4.4.5. in textbook.*) If X and Y are independent, then

$$\text{Cov}(X, Y) = 0 \quad \text{and hence} \quad \rho_{XY} = 0$$

(i) The converse is **not** true!

$$\text{Cov}(X, Y) = 0 \quad (\text{ or } \rho_{XY} = 0) \not\Rightarrow \text{independence.}$$

- $\text{Cov}(X, Y)$ measures only linear dependence. X and Y may not depend linearly (e.g., non-linear relationship or the space of Y may depend on X).
- The *worst jump*: "$\rho_{XY} = 0$ so then X does not *cause* Y". **NO. NO. NO!**
  - $Cov(X, Y) = 0$ does not imply independence
  - $Cov(X, Y) = 0$ is *only* about *linear* relationship
  - $Cov(X, Y)$ is *only* about an *average* relationship (about the corresponding mean of each variable) (say, X may still cause Y for each individual).

· Sadly, such a wrong jump in logic is made much more often than you can imagine :-(

(ii) If X and Y are independent, then (since $Cov(X, Y) = 0$):

$$\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y)$$

6. Two Useful Theorems

6.1. Law of Iterated Expectations

Theorem (Law of Iterated Expectations; LIE): *(Thm. 4.4.3. in textbook.)* Given two random variables X and Y:

$$E(Y) = E_X\left[E(Y|X)\right]$$

Notes on LIE:

- Example: Let Y = grade from an exam. Also, let $X = 1$ if female and $X = 0$ if male. Then, using the LIE we can get the average grade $E(Y)$ using the (conditional) average grade of females and males as:

$$E(Y) = E_X\left[E(Y|X)\right] = E(Y|X = 0) \cdot Pr(X = 0) + E(Y|X = 1) \cdot Pr(X = 1)$$

Or in words as

$$\begin{aligned}\text{Mean grade} = &(\text{Mean grade of males}) \cdot (\text{Fraction of males}) \\ &+ (\text{Mean grade of females}) \cdot (\text{Fraction of females})\end{aligned}$$

- Proof: See lecture. This is a good exercise on what we have covered in these notes (conditional, joint and marginal distributions, and expectations).

- Given a function $r(Y)$ of the random variable Y, the LIE can also be written as:

$$E(r(Y)) = E_X \left[E(r(Y)|X) \right]$$

- More generally, given a function $r(Y)$ of the random variable Y and two other random variables X and Z, we can write the conditional expectation of $r(Y)$ given Z as:

$$E[r(Y)|X] = E_{Z|X} \left[E[r(Y)|X, Z]|X \right]$$

- In practice (e.g., in books and papers), we usually do not specify what is the random variable in the outer expectation (it is assumed the reader knows). For example, we just write $E(Y) = E\left[E(Y|X)\right]$, or $E[r(Y)|X] = E\left[E[r(Y)|X, Z]|X\right]$.

6.2. Jensen's Inequality

Theorem (Jensen's Inequality): *(Thm. 4.7.7. in textbook.).* For any random variable X, if $g(X)$ is a *convex* function, then:

$$E[g(X)] \geq g(E[X])$$

and if $g(X)$ is a *concave* function, then:

$$E[g(X)] \leq g(E[X]),$$

with strict inequalities if the functions are strictly convex or concave.

Notes on Jensen's inequality:

- Example 1: Let $g(X) = X^2$, which is a strictly convex function. Then, by Jensen's inequality we have:

$$E[X^2] = E[g(X)] > g(E[X]) = (E[X])^2$$

- Example 2: Let $g(X) = \ln(X)$, which is a strictly concave function. Then, by Jensen's inequality we have:

$$E[\ln(X)] = E[g(X)] < g(E[X]) = \ln(E[X])$$

- This inequality can be useful to check whether an estimator is upward or downward biased (later).
- Graphical intuition of result: See lecture.

7. Multivariate Distributions

Idea: We can generalize all distributions discussed in the previous notes to many variables.

Here, we focus on a key one: Bivariate Normal Distribution

7.1. Bivariate Normal Distribution

Definition: *(Def. 4.5.10. in textbook.)*.(X, Y) follow a Bivariate Normal Distribution (BND) if their joint pdf is:

$$f(x,y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \cdot \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2 + \left(\frac{y-\mu_y}{\sigma_y}\right)^2 - 2\rho\left(\frac{x-\mu_x}{\sigma_x}\right)\left(\frac{y-\mu_y}{\sigma_y}\right)\right]\right)$$

for $(x,y) \in \mathbb{R}^2$, $\rho = \frac{\text{Cov}(X,Y)}{\sigma_X\sigma_Y}$, $\mu_x, \mu_y \in \mathbb{R}$, $\sigma_x, \sigma_y > 0$ and $\rho \in [-1,1]$

We usually write:

$$(X,Y) \sim N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$$

Notes on Bivariate Normal Distribution:

1. The marginals are normal distributions: $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$

   - Careful: The converse is **not** true!

   $$X \sim N(\mu_x, \sigma_x^2) \text{ and } Y \sim N(\mu_y, \sigma_y^2) \not\Rightarrow f_{xy}(X,Y) \text{ (joint pdf) is normal}$$

2. If $(X,Y) \sim N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$, then:

   $$\rho = 0 \textit{ if and only} \text{ if } X \text{ and } Y \text{ are independent.}$$

3. *Cool Property of the BND:* If $(X,Y) \sim N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$, then the conditional distribution of Y given X is:

   $$Y|X \sim N(\mu_{Y|X}, \sigma_{Y|X}^2), \text{ where}$$

   $$\mu_{Y|X} = E[Y|X] = \mu_y + \rho\frac{\sigma_y}{\sigma_x}(X - \mu_x) \quad \text{and} \quad \sigma_{Y|X}^2 = \text{Var}[Y|X] = \sigma_y^2(1-\rho^2)$$

Note that we can write the conditional mean $E(Y|X)$ as:

$$\begin{aligned}\mu_{Y|X} &= E[Y|X] \\ &= \mu_y + \rho\frac{\sigma_y}{\sigma_x}(X - \mu_x) \\ &= \left(\mu_y - \mu_x\rho\frac{\sigma_y}{\sigma_x}\right) + \rho\frac{\sigma_y}{\sigma_x}X \\ &= a + bX\end{aligned}$$

- This is just OLS!
- Hence, when (X, Y) follow jointly a bivariate normal distribution, the conditional expectation $E(Y|X)$ is *exactly* a *linear* function of X. This is important for linear regression analysis (or Ordinary Least Squares, OLS).
- Note that the slope coefficient equals:

$$b = \rho\frac{\sigma_y}{\sigma_x} = \frac{Cov(X,Y)}{\sigma_x\sigma_y}\frac{\sigma_y}{\sigma_x} = \frac{Cov(X,Y)}{Var(X)}$$

That is the formula of β in OLS! (More later, in GSE 520)

8. Extension to more than Two Random Variables

We can generalize everything we have done with two random variables to the case with *many* random variables: $(X_1, X_2, \ldots, X_n)$.

Examples:

- Joint pdf/pmf: $f_{X_1 \ldots X_n}(x_1, x_2, \ldots, x_n)$
- Marginal pdf of X_1: $f_{X_1}(x_1) = \int_{x_2}\int_{x_3}\cdots\int_{x_n} f_{X_1 \ldots X_n}(x_1, x_2, \ldots, x_n)\, dx_2 dx_3 \ldots dx_n$

- Joint pdf of (X_1, X_2): $f_{X_1X_2}(x_1, x_2) = \int_{x_3} \cdots \int_{x_n} f_{X_1 \ldots X_n}(x_1, x_2, \ldots, x_n)\, dx_3 \ldots dx_n$
- Conditional distributions:

$$f_{X_2 \ldots X_n | X_1}(x_2, \ldots, x_n | x_1) = \frac{f_{X_1 X_2 \ldots X_n}(x_1, x_2, \ldots, x_n)}{f_{X_1}(x_1)}$$

$$f_{X_3 \ldots X_n | X_1, X_2}(x_3, \ldots, x_n | x_1, x_2) = \frac{f_{X_1 X_2 \ldots X_n}(x_1, x_2, \ldots, x_n)}{f_{X_1 X_2}(x_1, x_2)}$$

- Independence: $f_{X_1X_2 \ldots X_n}(x_1, x_2, \ldots, x_n) = f_{X_1}(x_1) \cdot f_{X_2}(x_2) \cdots f_{X_n}(x_n)$